

**KENYATTA UNIVERSITY**  
**UNIVERSITY EXAMINATIONS 2019/2020**

**SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE**  
**IN MECHANICAL ENGINEERING**

**EMM 101: ENGINEERING DRAWING AND DESIGN II**

**DATE: WEDNESDAY 16<sup>TH</sup> DECEMBER 2020**

**TIME: 11.00 A.M. - 1.00 P.M.**

**INSTRUCTION**

- 1) This paper contains (5) Five questions.
- 2) Quest (1) One is compulsory and carries 30 marks.
- 3) Answer any (2) Two questions from the remaining (4) Four.
- 4) The Four questions carry 20 marks each.

**Q1.** The part shown in figure (Fig) Q1 is a locking device used to secure a shaft to a bracket. Draw in 3<sup>rd</sup> angle projection the following:

- |    |                                                            |           |
|----|------------------------------------------------------------|-----------|
| a) | A <u>front elevation</u> as obtained from the direction A. | (6 marks) |
| b) | An <u>end view</u> obtained from the direction B.          | (6 marks) |
| c) | A <u>plan view</u>                                         | (6 marks) |
| d) | Insert major dimensions                                    | (5 marks) |
| e) | Show the projection symbol                                 | (2 marks) |
| f) | A suitable title block and borders                         | (5 marks) |

**Q2.** a) Fig 2A shows an orthographic view of a T.V. stand accessory.  
Draw an isometric view of the bracket with the corner A as the lowest point on the drawing.

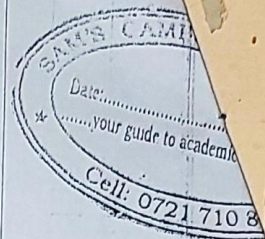
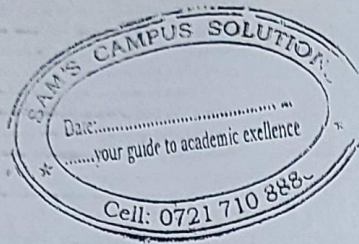
(10 marks)

b) Fig 2B shows the orthographic view of an engineering object.  
Draw an oblique view of the object in cavalier projection using a receding angle 30° to the right.

(10 marks)

Note: For questions 2a and 2b use a scale of 1:2



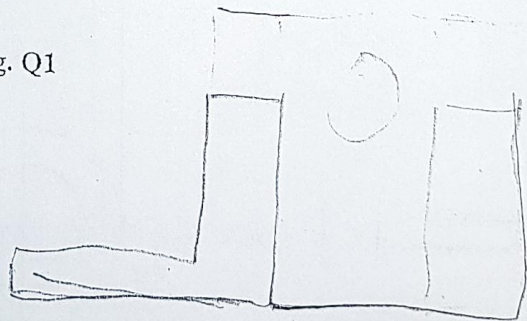
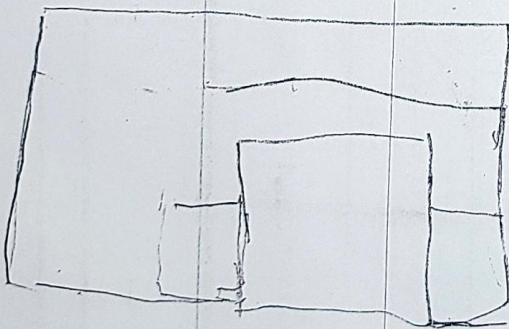
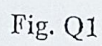


- Q3. a) A rectangle measuring 50 mm length and 30 mm width has limits  $\pm 0.4$  mm. Clearly illustrate the toleranced drawing and interpretation. (5 marks)
- b) Using Data sheet BS4500A, provide the ISO fits and tolerance, and calculate the maximum and minimum limits for the following hole and shaft nominal sizes.
- i) 30 H10/c11
  - ii) 18H7/n6
  - iii) 1SOH7/s6
- (6 marks)
- c) With the aid of a drawing, illustrate the three types of fit. (9 marks)

Q4. Fig Q4 shows three views of an object in first angle projection. Draw full isometric sectional view from the end. (20 marks)

- Q5. a) Draw a circuit in Fig 5a, using correct graphical symbols in place of the boxes. (4 marks)
- b) Redraw the circuit diagram with graphical symbols twice:
- i) With an indicator in series with the motor.
  - ii) With an indicator in parallel with the motor.
- (8 marks)
- c) Fig Q5b shows a double-line drawing for a system of pipe fittings. Redraw the figure as a single-line drawing and label all the fittings. (8 marks)







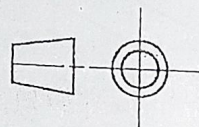
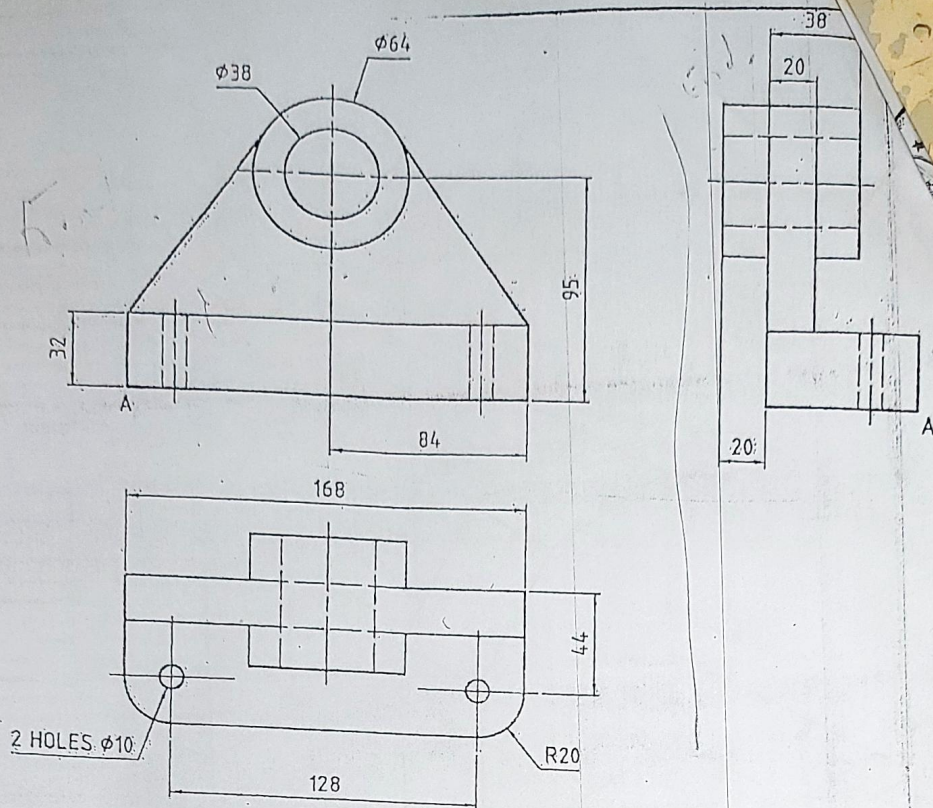


Fig 2A

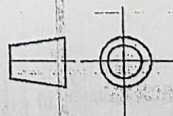
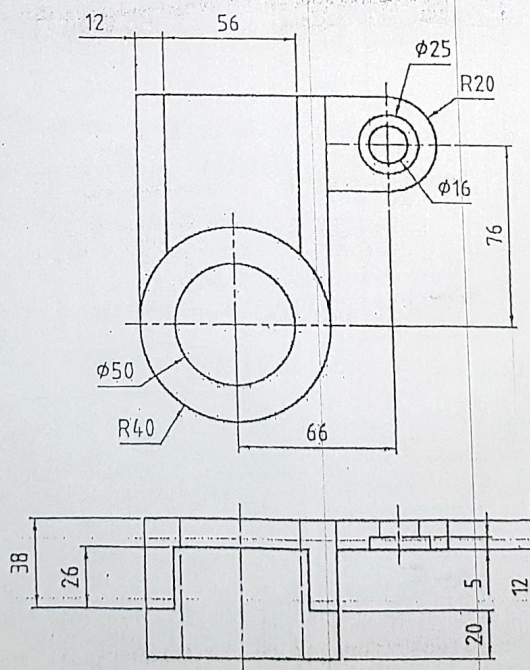


Fig 2B



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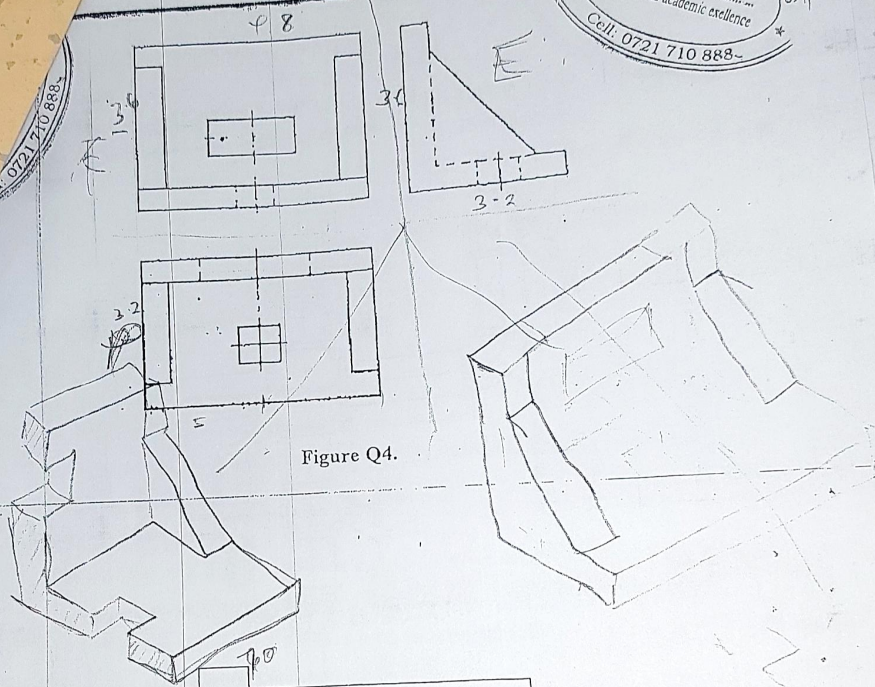
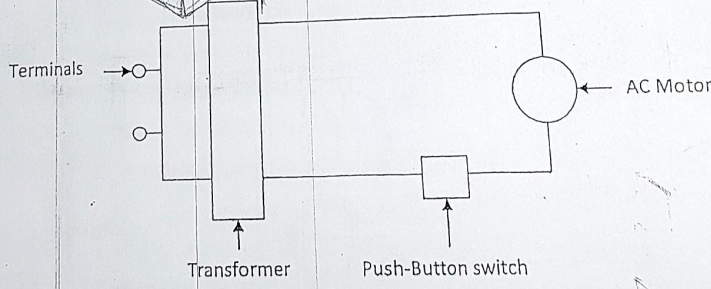


Figure Q4.





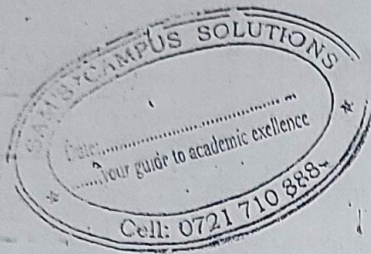


Fig Q5b) Pipe and pipe fittings

